Develop a program to perform Bias Audit on pre-trained word embeddings to identify gender or racial stereotypes using the Word Embedding Association Test (WEAT).

Dataset: Common Crawl or Google News Embeddings.

Code:

```
pip install gensim numpy scipy
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ---------------------------------------- 27.9/27.9 MB 34.8 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```python
import gensim.downloader as api

model = api.load("word2vec-google-news-300")
```

```
[==================================================] 100.0% 1662.8/1662.8MB downloaded
```

```python
male_terms = [
    "man","male","boy","brother","he","him","son"
]

female_terms = [
    "woman","female","girl","sister","she","her","daughter"
]

career_words = [
    "career","corporation","salary","office","professional"
]

family_words = [
    "home","parents","children","family","cousins"
]
```

```python
import numpy as np

def cosine_similarity(a,b):
    return np.dot(a,b)/(np.linalg.norm(a)*np.linalg.norm(b))
```

```python
def association(word, A, B, model):

    simA = np.mean([cosine_similarity(model[word], model[a]) for a in A])
    simB = np.mean([cosine_similarity(model[word], model[b]) for b in B])

    return simA - simB
```

```python
def weat_score(X,Y,A,B,model):

    score_X = sum([association(x,A,B,model) for x in X])
    score_Y = sum([association(y,A,B,model) for y in Y])

    return score_X - score_Y
```

```python
X = male_terms
Y = female_terms
A = career_words
B = family_words

score = weat_score(X,Y,A,B,model)

print("WEAT Bias Score:",score)
```

Output:

```
WEAT Bias Score: 0.24307287
```